

ESE5180: Lab 0 Zephyr

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GitHub Repository URL: https://github.com/sydfitz/ese5180_lab0

1. Hello (Vanilla) Zephyr

(1.1) Done; video uploaded to Google form

2. Hello (Nordic) Zephyr

(2.1) Done; Zephyr application committed under blinky_revised/.

(2.2) Done; video uploaded to Google Form

3. Building with West

(3.1) Terminal output:

```
Problems Output Debug Console Terminal Ports powershell
PS C:\Users\sydney\26-27\ESE_5180\ese5180_lab0> west build -p always -b nrf7002dk/nrf5340/cpuapp/ns -d blinky/build blinky --sysbuild
-- west build: making build dir C:\Users\sydney\26-27\ESE_5180\ese5180_lab0\blinky\build pristine
-- west build: generating a build system
Loading Zephyr module(s) (Zephyr base): sysbuild_default
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum required is "3.12") found components: Interpreter
-- Cache files will be written to: C:/ncs/v3.4.0/zephyr/.cache
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: nrf7002dk, qualifiers: nrf5340/cpuapp/ns
Parsing C:/ncs/v3.4.0/zephyr/share/sysbuild/Kconfig
Loaded configuration 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/_sysbuild/empty.conf'
Merged configuration 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/_sysbuild/empty.conf'
Configuration saved to 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/zephyr/.config'
Kconfig header saved to 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/_sysbuild/autoconf.h'
--
*****
* Running CMake for blinky *
*****

Loading Zephyr default modules (Zephyr base).
-- Application: C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky
-- CMake version: 4.2.1
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum required is "3.12") found components: Interpreter
-- Cache files will be written to: C:/ncs/v3.4.0/zephyr/.cache
-- Zephyr version: 4.4.0 (C:/ncs/v3.4.0/zephyr)
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: nrf7002dk, qualifiers: nrf5340/cpuapp/ns
-- Found host-tools: zephyr 1.0.1 (C:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk)
-- Found toolchain: zephyr 1.0.1 (C:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk)
-- Found Dtc: C:/ncs/toolchains/dcbdc366a1/opt/bin/dtc.exe (found suitable version "1.4.7", minimum required is "1.4.6")
-- Found BOARD.dts: C:/ncs/v3.4.0/zephyr/boards/nordic/nrf7002dk/nrf7002dk_nrf5340_cpuapp_ns.dts
-- Generated zephyr.dts: C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/zephyr.dts
-- Generated pickled edt: C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/edt.pickle
-- Generated devicetree_generated.h: C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/include/generated/zephyr/devicetree_generated.h

warning: Experimental symbol TFM_EXPERIMENTAL is enabled.

Parsing C:/ncs/v3.4.0/zephyr/Kconfig
Loaded configuration 'C:/ncs/v3.4.0/zephyr/boards/nordic/nrf7002dk/nrf7002dk_nrf5340_cpuapp_ns_defconfig'
Merged configuration 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/prj.conf'
Merged configuration 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/.config.sysbuild'
Configuration saved to 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/.config'
Kconfig header saved to 'C:/Users/sydney/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/include/generated/zephyr/autoconf.h'
-- Found GnuLD: C:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk/gnu/arm-zephyr-eabi/arm-zephyr-eabi/bin/ld.bfd.exe (found version "2.43.1")
-- The C compiler identification is GNU 14.3.0
-- The CXX compiler identification is GNU 14.3.0
-- The ASM compiler identification is GNU
-- Found assembler: C:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk/gnu/arm-zephyr-eabi/bin/arm-zephyr-eabi-gcc.exe
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
-- Could NOT find Threads (missing: Threads_FOUND)
-- Performing Test C_COMPILER_SUPPORTS_WFORMAT_SIGNEDNESS
-- Performing Test C_COMPILER_SUPPORTS_WFORMAT_SIGNEDNESS - Success
-- Using ccache: C:/ncs/toolchains/dcbdc366a1/opt/bin/ccache.exe
-- Found gen_kobject_list: C:/ncs/v3.4.0/zephyr/scripts/build/gen_kobject_list.py
-- Configuring done (11.2s)
-- Generating done (0.9s)
```

```
Problems Output Debug Console Terminal Ports
-- Build files have been written to: C:/Users/sydne/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky
-- Configuring done (16.2s)
-- Generating done (0.1s)
-- Build files have been written to: C:/Users/sydne/26-27/ESE_5180/ese5180_lab0/blinky/build
-- west build: building application
[4/259] Generating include/generated/zephyr/version.h
-- Zephyr version: 4.4.0 (C:/ncs/v3.4.0/zephyr), build: ncs-v3.4.0
[10/259] Generating ../tfm/CMakeCache.txt
-- Found Git: C:/ncs/toolchains/dcbdc366a1/mingw64/bin/git.exe (found version "2.52.0.windows.1")
-- The C compiler identification is GNU 14.3.0
-- The CXX compiler identification is GNU 14.3.0
-- The ASM compiler identification is GNU
-- Found assembler: C:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk/gnu/arm-zephyr-eabi/bin/arm-zephyr-eabi-gcc.exe
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum required is "3.12") found components: Interpreter
-- Cache files will be written to: C:/ncs/v3.4.0/zephyr/.cache
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
CMake Warning (dev) at C:/ncs/v3.4.0/nrf/modules/trusted-firmware-m/tfm_boards/nrf5340_cpuapp/CMakeLists.txt:21 (install):
Policy CMP0177 is not set: install() DESTINATION paths are normalized. Run
  "cmake --help-policy CMP0177" for policy details. Use the cmake_policy
  command to set the policy and suppress this warning.
This warning is for project developers. Use -Wno-dev to suppress it.

-- Configuring done (2.1s)
-- Generating done (0.3s)
CMake Warning:
Manually-specified variables were not used by the project:

  QCBOR_PATH
  TFM_CREATE_SIGNING_LAYOUT_OBJECTS
  T_COSE_PATH

-- Build files have been written to: C:/Users/sydne/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/tfm
[193/195] Linking C executable bin\tfm.s.axf
Memory region      Used Size  Region Size  %age Used
  FLASH:           81364 B      256 KB      31.04%
    RAM:           45960 B      256 KB      17.53%
[195/195] Linking C static library secure_fw\libtfm_s_veneurs.a

[17/259] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[259/259] Linking C executable zephyr\zephyr.elf
Memory region      Used Size  Region Size  %age Used
  FLASH:           22576 B      192 KB      11.48%
    RAM:            4600 B      192 KB       2.34%
  IDT_LIST:         0 B        32 KB       0.00%
Generating files from C:/Users/sydne/26-27/ESE_5180/ese5180_lab0/blinky/build/blinky/zephyr/zephyr.elf for board: nrf7002dk/nrf5340/cpuapp/ns

[8/8] Completed 'blinky'
PS C:\Users\sydne\26-27\ESE_5180\ese5180_lab0> west flash -d blinky/build --erase
-- west flash: rebuilding
[0/16] Performing build step for 'tfm'ninja: no work to do.
```

```
-- west flash: rebuilding
[0/16] Performing build step for 'tfm'ninja: no work to do.

[2/3] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[3/3] Completed 'tfm'

[3/3] Completed 'blinky'
-- west flash: using runner nrfutil
-- runners.nrfutil: mass erase requested
-- runners.nrfutil: reset after flashing requested
Using board 001050764169
-- runners.nrfutil: Flashing file: C:\Users\sydne\26-27\ESE_5180\ese5180_lab0\blinky\build\blinky\zephyr\tfm_merged.hex
-- runners.nrfutil: Connecting to probe
-- runners.nrfutil: Erasing non-volatile memory (ERASEALL)
-- runners.nrfutil: Programming image
-- runners.nrfutil: Verifying image
-- runners.nrfutil: Reset
-- runners.nrfutil: Board(s) with serial number(s) 1050764169 flashed successfully.
PS C:\Users\sydne\26-27\ESE_5180\ese5180_lab0>
```

5. Device Tree (DT)

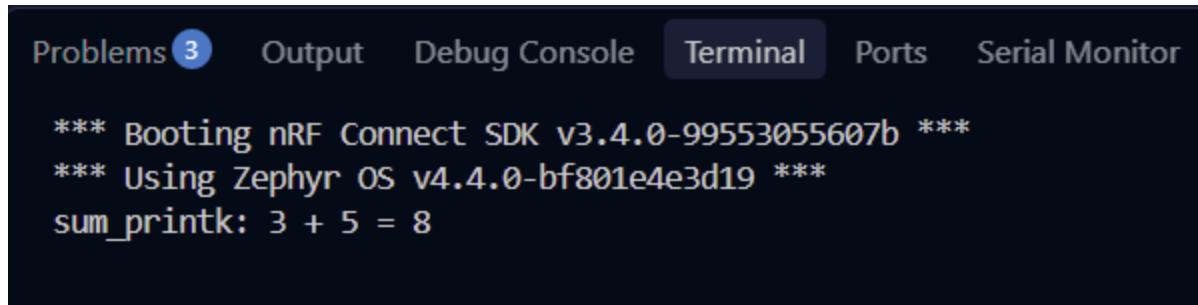
(5.1) Done; new alias defined in blinky_revised\boards\nrf7002dk_nrf5340_cpuapp_ns.overlay, used in blinky_revised\src\main.c.

(5.2) Done; button is polled every 50ms in the main loop in blinky_revised\src\main.c, toggles the led5180 on each new button press.

(5.3) Done; new button alias button5180 defined in
blinky_revised\boards\nrf7002dk_nrf5340_cpuapp_ns.overlay and
blinky_revised\boards\nrf7002dk_nrf5340_cpuapp.overlay, used in blinky_revised\src\main.c.

6. Printing vs. Logging

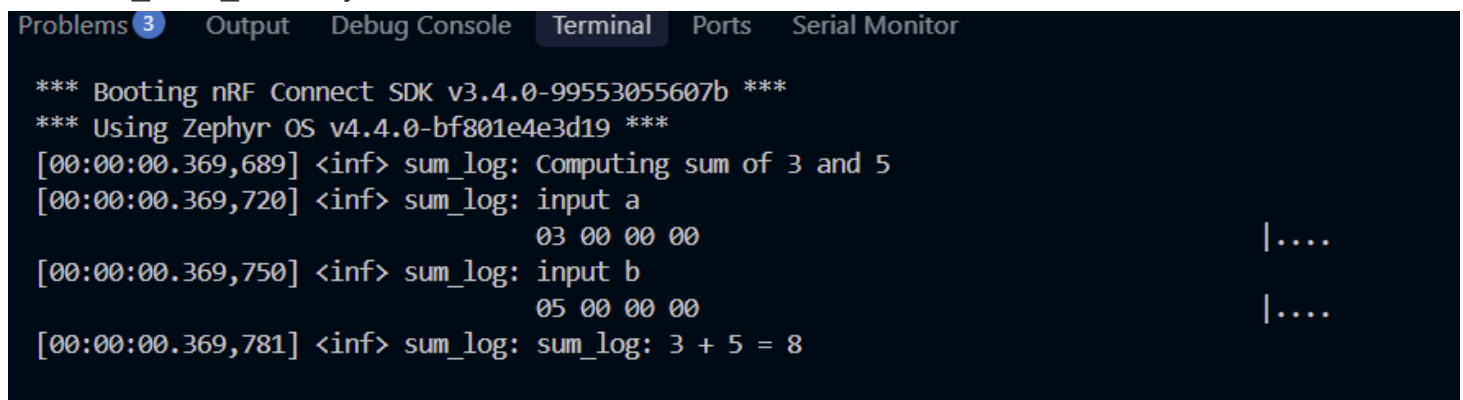
(6.1) CONFIG_SUM_PRINT = y:



```
Problems 3 Output Debug Console Terminal Ports Serial Monitor

*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
sum_printk: 3 + 5 = 8
```

CONFIG_SUM_LOG = y:



```
Problems 3 Output Debug Console Terminal Ports Serial Monitor

*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
[00:00:00.369,689] <inf> sum_log: Computing sum of 3 and 5
[00:00:00.369,720] <inf> sum_log: input a
                        03 00 00 00 | ....
[00:00:00.369,750] <inf> sum_log: input b
                        05 00 00 00 | ....
[00:00:00.369,781] <inf> sum_log: sum_log: 3 + 5 = 8
```

(6.2) Done; video upload to google form

(6.3) Done; the updated code for this part is in a new application folder named /print_log.

7. Ztest for Unit Testing

(7.1) Done; relevant Ztest code is under print_log/tests/SUM_UNIT_TEST

(7.2) Screenshot of testing outputs (running on qemu, I had to redirect the output path to C:\twout due to Windows path too long errors):

```

PS C:\Users\sydne\ESE_5180\ese5180_lab0\print_log\tests> west twister -T SUM_UNIT_TEST -p qemu_cortex_m3 --inline-logs -v -O /c/twout
Renaming previous output directory to /c/twout.1
INFO - Using Ninja..
INFO - Zephyr version: ncs-v3.4.0
INFO - Using 'zephyr/gnu' toolchain variant.
INFO - Building initial testsuite list...
INFO - Built testsuite list in 0.00 seconds
INFO - Writing JSON report /c/twout/testplan.json
INFO - Adding tasks to the queue...
INFO - Added initial list of jobs to queue
INFO - 1/1 qemu_cortex_m3/ti_lm3s6965 sum_log.test                                     <[32mPASSED<[39m (qemu 7.960s <zephyr/gnu>)

INFO - 1 test scenarios (1 configurations) selected, 0 configurations filtered (0 by static filter, 0 at runtime).
INFO - <[32m1 of 1<[39m executed test configurations passed (100.00%), 0 built (not run), 0 failed, 0 errored, with no warnings in 35.50 seconds.
INFO - 3 of 3 executed test cases passed (100.00%) on 1 out of total 1514 platforms (0.07%).
INFO - <[32m1<[39m test configurations executed on platforms, <[36m0<[39m test configurations were only built.
INFO - Saving reports...
INFO - Writing JSON report C:\c\twout\twister.json
INFO - Writing xunit report C:\c\twout\twister.xml...
INFO - Writing xunit report C:\c\twout\twister_report.xml...
INFO - Run completed
PS C:\Users\sydne\ESE_5180\ese5180_lab0\print_log\tests>

```

8. Adding a Peripheral (BME280)

(8.1) Done; temp printing code is under print_log/bme280_temp/. CONFIG_TEMP_READ is a separate Kconfig option from the CONFIG_SUM_PRINT / CONFIG_SUM_LOG choice.

```

BME280 temperature: 22.80 C
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
sum_printk: 3 + 5 = 8
BME280 temperature: 22.87 C

```

(8.2) Done; Ztest code is under print_log/tests/BME280_DT_TEST, testing that the BME280 devicetree node/I2C bus are set up correctly and sanity-checking sensor. I had the same file path length issue, so I built to a shorter output location than the default.

```
bme280 temperature: 22.87 C
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE bme280_dt_test_suite
```

```
=====
START - test_bme280_compatible
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:20 (bme280_dt_test_suite_test_bme280_compatible)
PASS - test_bme280_compatible in 0.011 seconds
=====
```

```
START - test_bme280_i2c_address
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:26 (bme280_dt_test_suite_test_bme280_i2c_address)
PASS - test_bme280_i2c_address in 0.011 seconds
=====
```

```
START - test_bme280_i2c_bus_ready
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:32 (bme280_dt_test_suite_test_bme280_i2c_bus_ready)
PASS - test_bme280_i2c_bus_ready in 0.011 seconds
=====
```

```
START - test_bme280_node_enabled
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:14 (bme280_dt_test_suite_test_bme280_node_enabled)
PASS - test_bme280_node_enabled in 0.011 seconds
=====
```

```
START - test_compensate_temperature_known_value
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:55 (bme280_dt_test_suite_test_compensate_temperature_known_value)
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:56 (bme280_dt_test_suite_test_compensate_temperature_known_value)
PASS - test_compensate_temperature_known_value in 0.024 seconds
=====
```

```
START - test_compensate_temperature_zero_coeffs
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:46 (bme280_dt_test_suite_test_compensate_temperature_zero_coeffs)
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:47 (bme280_dt_test_suite_test_compensate_temperature_zero_coeffs)
PASS - test_compensate temperature zero coeffs in 0.024 seconds
=====
```

```
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_dt_test.c:47 (bme280_dt_test_suite_test_compensate_temperature_zero_coeffs)
PASS - test_compensate temperature zero coeffs in 0.024 seconds
=====
```

```
TESTSUITE bme280_dt_test_suite succeeded
```

```
----- TESTSUITE SUMMARY START -----
```

```
SUITE PASS - 100.00% [bme280_dt_test_suite]: pass = 6, fail = 0, skip = 0, total = 6 duration = 0.092 seconds
```

- PASS - [bme280_dt_test_suite.test_bme280_compatible] duration = 0.011 seconds
- PASS - [bme280_dt_test_suite.test_bme280_i2c_address] duration = 0.011 seconds
- PASS - [bme280_dt_test_suite.test_bme280_i2c_bus_ready] duration = 0.011 seconds
- PASS - [bme280_dt_test_suite.test_bme280_node_enabled] duration = 0.011 seconds
- PASS - [bme280_dt_test_suite.test_compensate_temperature_known_value] duration = 0.024 seconds
- PASS - [bme280_dt_test_suite.test_compensate_temperature_zero_coeffs] duration = 0.024 seconds

```
----- TESTSUITE SUMMARY END -----
```

```
=====
PROJECT EXECUTION SUCCESSFUL
```